SERVER Project



8953 - Open Adapter Card (OAC) Sub-Rackscale Acceleration Lives On

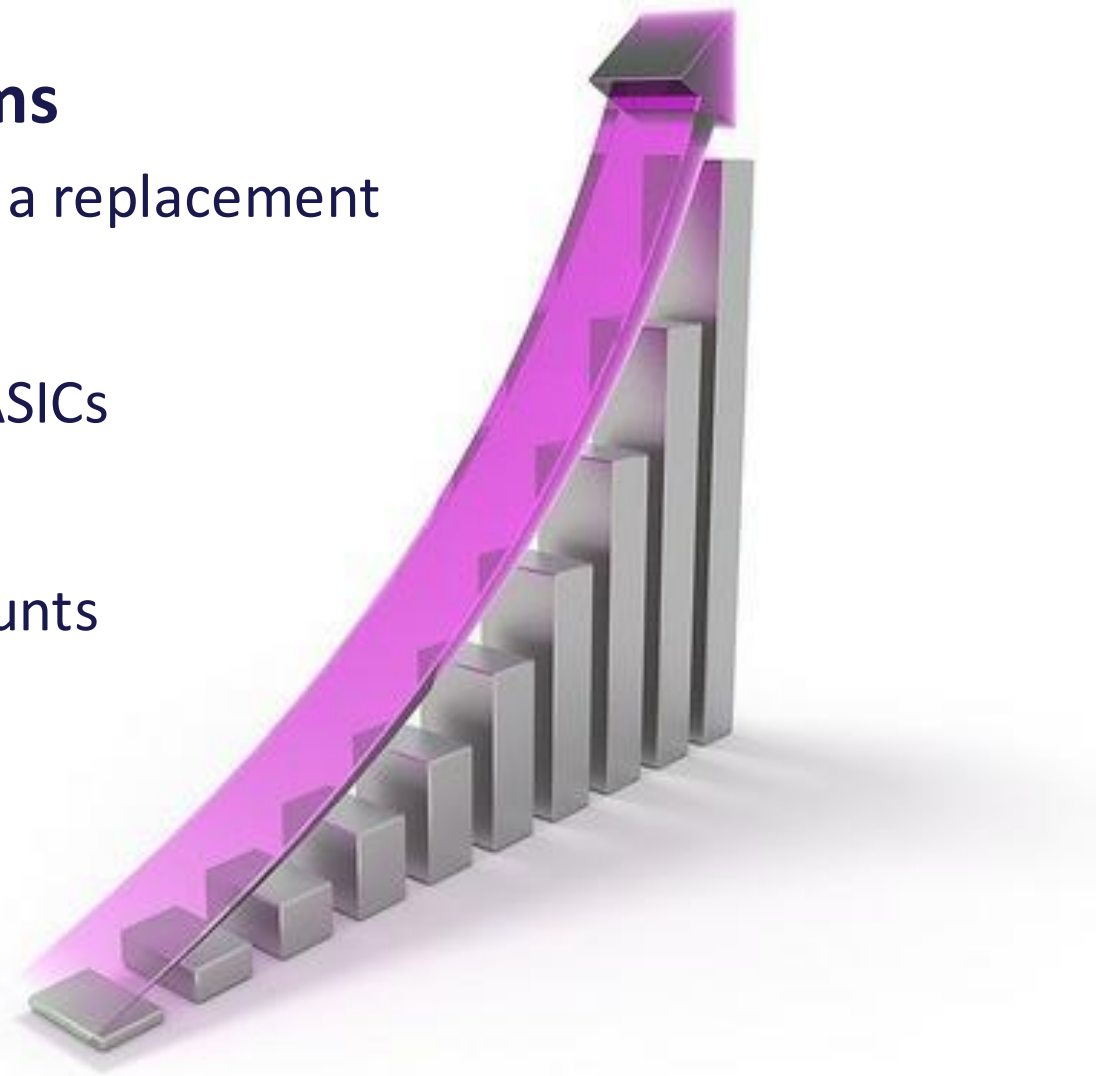
Greg Sellman – *PMTS – Advanced Micro Devices (AMD)*

Brian Aspnes – *Senior Principal Hardware Architect – Intel Corporation*

OCP PLATINUM Member

Why OAC, Why Now?

- **Must Continue to Address Pod of 8 Platforms**
  - OAI/OAM was sunset in 2024; industry requires a replacement
- **Larger Silicon**
  - More cores & more memory are driving larger ASICs
- **Scale-up & Scale-Out Requirements**
  - Need for agnostic connectivity & higher lane counts
- **2x-8x Increase in Power**
  - More cores, lanes & memory = More Power
- **Advances in Cooling**
  - Emergence and widespread adoption of DLC



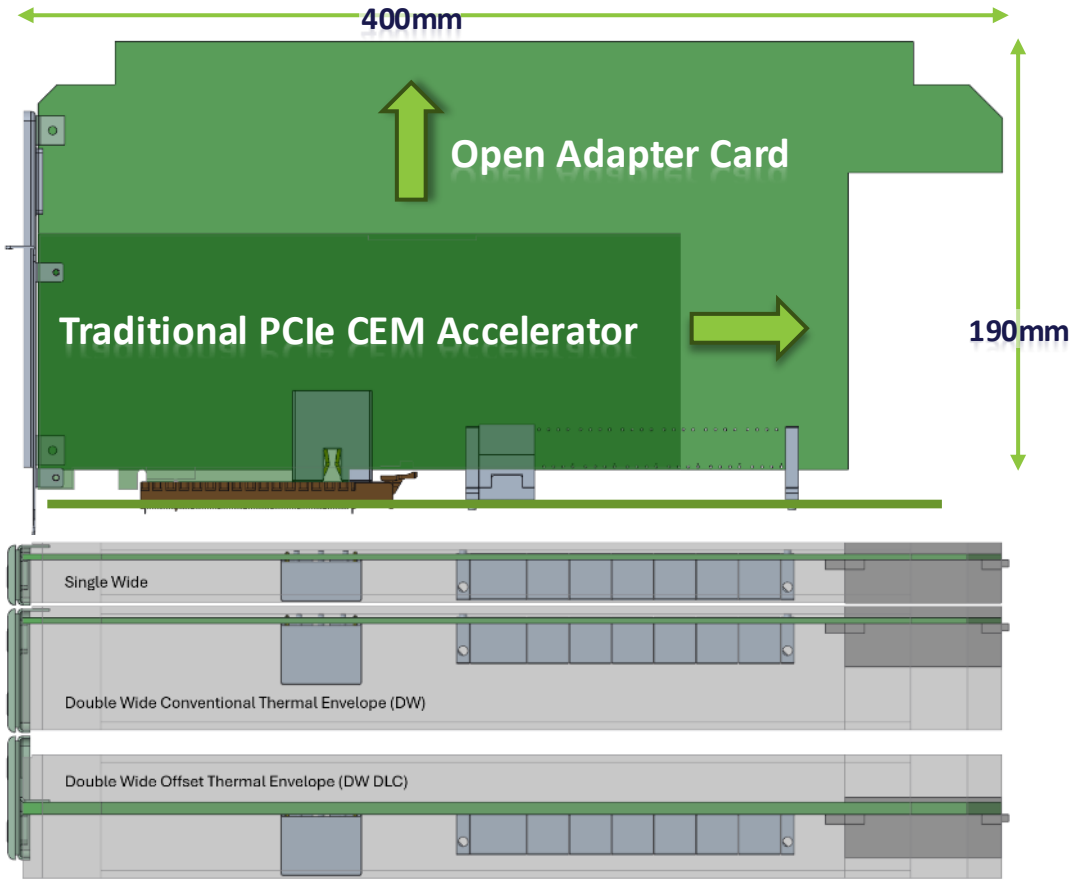
Open Adapter Card (OAC)

Cliffs Notes

One Form Factor, Multiple Platforms
Enables common chassis, cooling, power delivery, and serviceability architectures across multiple adapter vendors.

One Form Factor, Multiple Fabrics
Supports a diverse selection of high-speed protocols and fabrics within a standardized mechanical and electrical envelope.

Reduced Platform Development Cost
Minimizes platform redesign by decoupling system infrastructure decisions from adapter vendor selection.



Up to 4.5x Increase in Usable PCB Area

- PCIe CEM to Double-wide Offset OAC
- **3.2x Increase in Vol.;** DW PCIe CEM to DW OAC

112L of 200Gbps HSIO

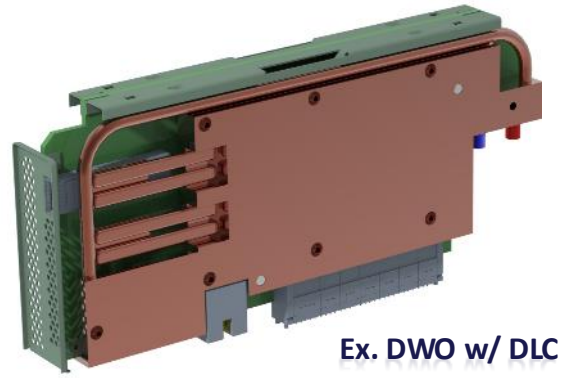
- New ZettaMAX™ 2-pc. Connector Interface

5000W Capable

- 5000W max @ 50V / 1200W max @ 12V
- Single-wide 4000W / 960W max

DLC or Air-Cooled

- MQDB-04 QD Defined Interface for DLC



Ex. DWO w/ DLC

Three Width Profiles **support up to 16x cards within a 19" chassis**

Single Wide - SW

17.5mm Front Envelope
7mm (max) Rear Envelope

Double Wide - DW

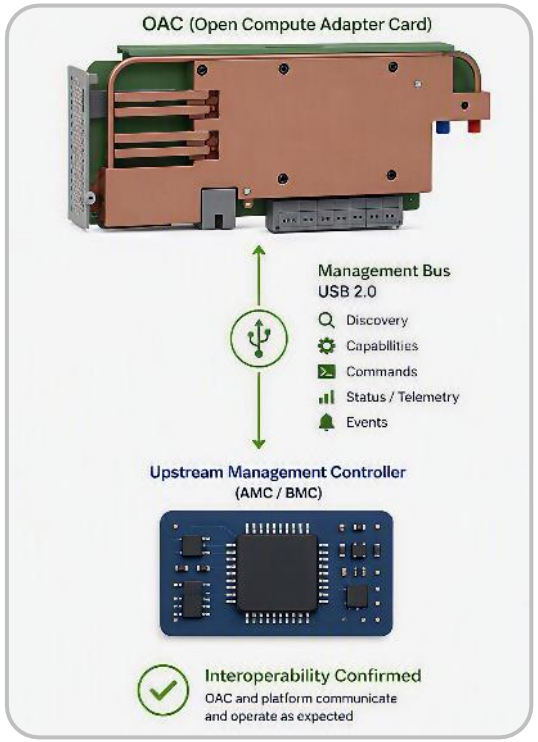
43.5 mm Front Envelope
7mm (max) Rear Envelope

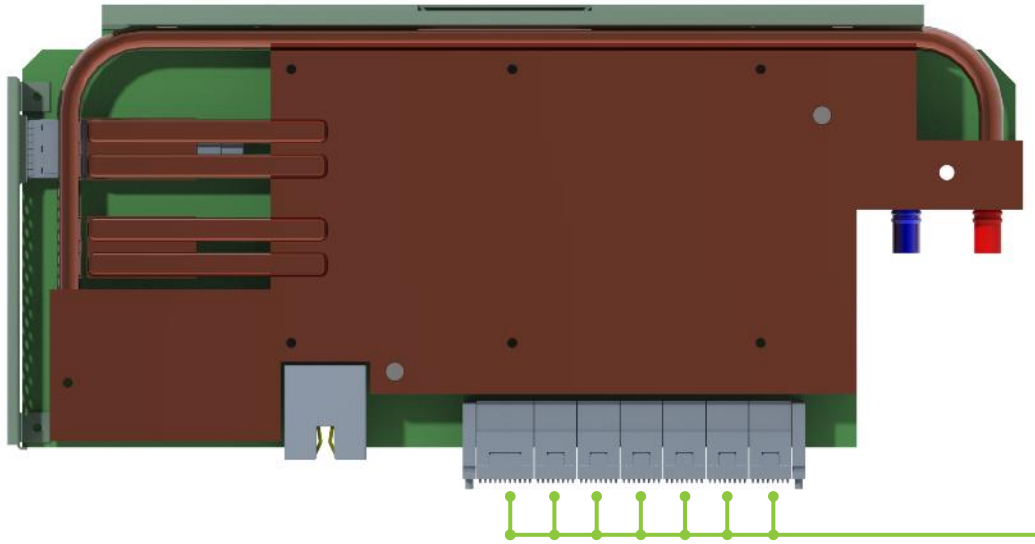
Double Wide Offset - DWO

26mm Front Envelope
24.5mm (max) Rear Envelope

Open Adapter Card (OAC)

Protocol-Agnostic by Design





7 **High-speed Connectors**
Up to 224 differential pairs

PCIe Gen 7
High-performance bandwidth

200G Ethernet
High throughput networking

NVLink
High speed GPU-to-GPU interconnect

UALink
Open, scalable interconnect

CXL
Efficiency memory expansion

Future Interconnects
Ready for what's next

Enabled by two *NEW* architectural concepts

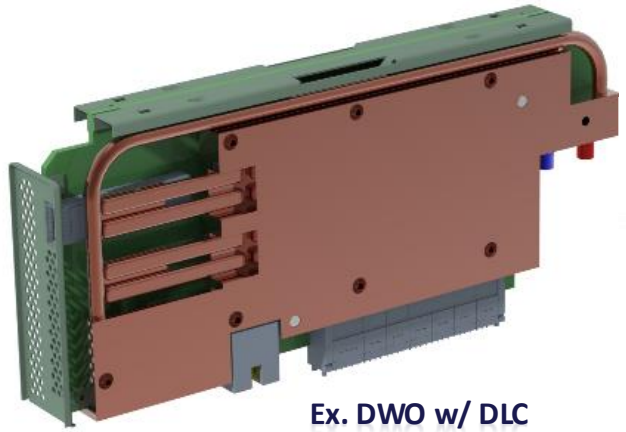
- Connector Interface Definition (CID)
- Slot Configuration Data (SCD)

Adapter Vendor Defined

Up to **224** Differential Pairs

Up to **50** Discrete Signals

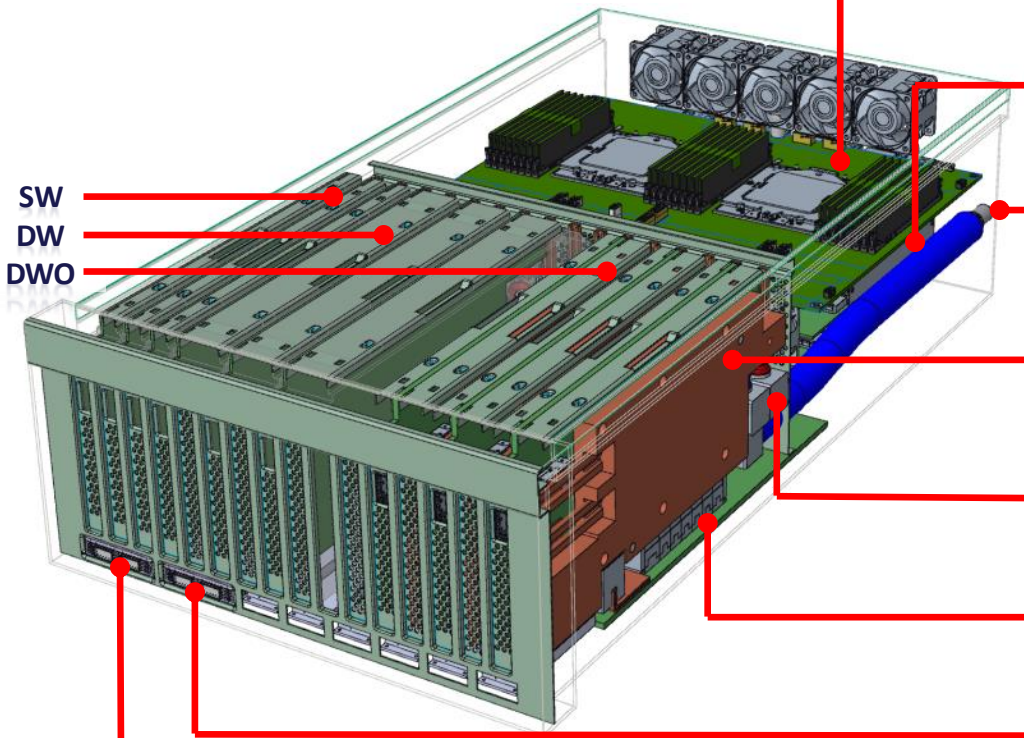
Open, Modular Platform Enablement - DLC



Ex. DWO w/ DLC

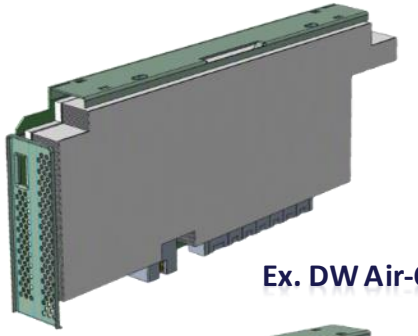
19” Rack Optimized, Open Rack Compatible
- Up to 16x SW adapters in 19”
- Up to 8x DW or DWO adapters in 19”

Example 5RU Platform

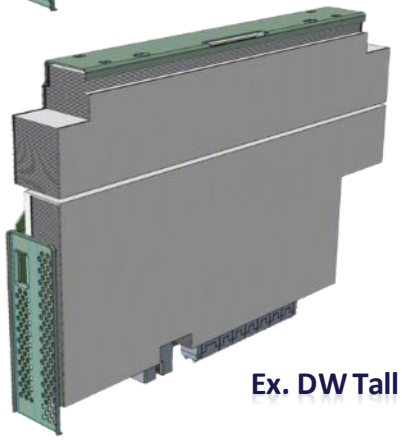


- **OCP DC-MHS HPM**
- **OCP M-CRPS PSU**
- **OCP UQD v2.0**
- **OCP Open Adapter Card**
  Standard Height, Double-wide Offset w/ DLC
- **OCP MQD-B 04**
- **Modular Baseboard**
- **OCP NIC 3.0**
- **OCP DC-SCM 2.1**

Open, Modular Platform Enablement – Air Cooled



Ex. DW Air-Cooled

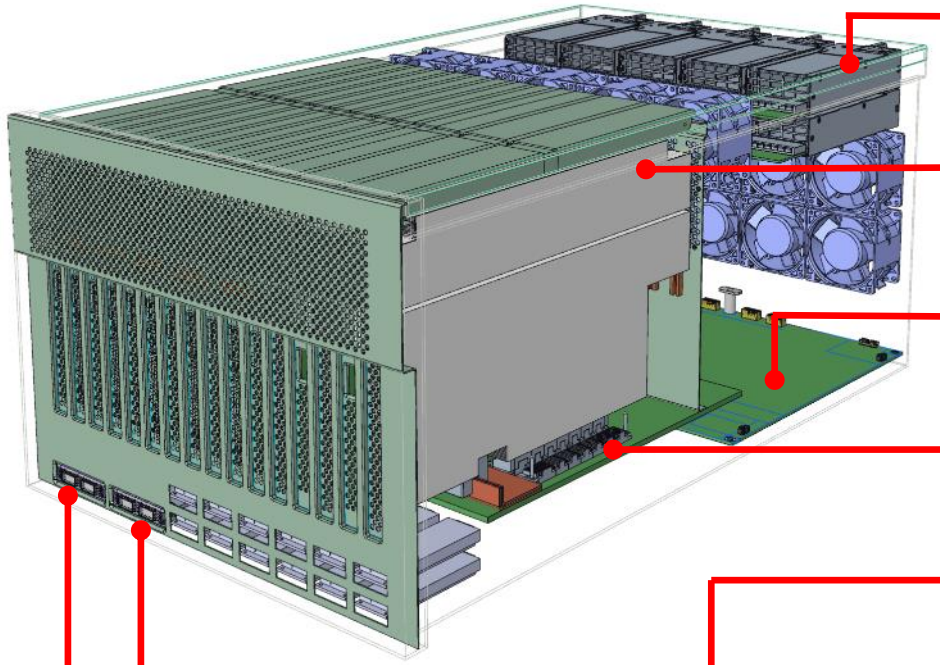


Ex. DW Tall Air-Cooled

Air Cooling Options Remain; Enterprise Demands It

- Up to ~700W in Double-Wide
- Up to ~1200W in Double-Wide Tall

Example 8RU Platform



- **OCP M-CRPS PSU**
- **OCP Open Adapter Card**
  Tall Height, Double-wide Air Cooled
- **OCP DC-MHS HPM**
- **Modular Baseboard**
- **OCP NIC 3.0**
- **OCP DC-SCM 2.1**

Call To Action

- Support for open, modular platforms continues to drive innovation and new ideas. Get involved, provide feedback, help to grow an OAC ecosystem.

- Specification Timeline
  - v0.9 - Available Today
  - https://www.opencompute.org/w/index.php?title=Server/OpenAdapterCard

- Provide feedback
  - AMD – Greg Sellman – gsellman@amd.com
  - Qualcomm – Jared Wright – jared.wright@qualcomm.com
  - Dell – Corey Hartman – corey.hartman@dell.com
  - Intel – Brian Aspnes – brian.d.aspnes@intel.com
  - Nvidia – John Norton – johnn@nvidia.com



Thank You!

